

AAYUSH CHOPRA

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EDUCATION

Master of Science in Computer and Information Science

Sept 2024 - April 2026

University of Michigan – Dearborn

Coursework: Artificial Intelligence, Computer Network & Security, Robot Vision, Wireless Technology, Advance Operating System, Edge Computing

Bachelor of Technology in Computer Science Engineering

July 2016 - June 2019

Kurukshetra University, Kurukshetra, India

EXPERIENCE

Ongraph Technology Pvt. Ltd. Noida, India

Jan 2020 - Aug 2024

Software Engineer

- Refactored a monolithic application into 10+ microservices, engineering scalable RESTful APIs and improving system efficiency, resulting in a 60% increase in scalability and a 3x boost in deployment frequency.
- Developed cloud-native solutions and real-time dashboard systems, reducing data retrieval time by 90% and accelerating business decision-making by 50%.
- Integrated third-party APIs including SAML SSO and Health Hero to enable seamless banking and virtual healthcare services, enhancing user experience and reducing page load time by 40%.
- Architected and automated CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment time by 40%, and streamlined agile processes with Jira, improving sprint efficiency by 30%.
- Validated critical user workflows, including multi-role account registration, product search, payment processing, and order confirmation, ensuring system functionality and enhancing user experience.
- Cultivated strong client relationships and collaborated with development teams, achieving a 95% on-time project delivery rate and a 30% increase in client satisfaction.
- Facilitated Agile ceremonies sprint planning, retrospectives, and daily standups to strengthen QA collaboration and boost test coverage by 35% across multiple projects
- Implemented secure payment gateways like Stripe and GoCardless, reducing transaction failure rates by 20% and enabling seamless reliable transaction processing.

PROJECTS

Real-Time Crowd Behavior Analysis for Risk Mitigation | Python, OpenCV, TensorFlow, YOLO, Real-Time Video Processing

- Introduced the system on edge devices (Raspberry Pi) for local processing, reducing average latency by 40% and enabling rapid response in large venues with over 500+ people
- Developed a basic alerting system that detects and flags crowd density spikes exceeding 75% of safe occupancy levels.

IoT-Based Real-Time Monitoring System | AWS IoT Core, Lambda, API Gateway, EC2, SNS, MQTT Broker, Raspberry Pi, Python

- Achieved a 30% reduction in system latency by optimizing MQTT communication and leveraging AWS IoT Core for efficient device-to-cloud data transmission.
- Leveraged serverless AWS services (Lambda, API Gateway) to minimize infrastructure costs, reducing operational overhead by 20% compared to traditional server-based solutions.

Speech Recognition Android App | Google Cloud Speech-to-Text API, gRPC, Java, Android Studio, REST

- Reduced transcription latency by over 25% by integrating gRPC-based streaming and optimizing Protobuf message handling.
- Designed a modular architecture with clear separation between audio pre-processing, API interaction, and UI, improving maintainability and feature extensibility.

TECHNICAL SKILLS

- Programming Languages: Python, Ruby on Rails, MATLAB, C++, JavaScript (React)
- Backend Development: REST/gRPC APIs, Microservices
- Database & Storage: PostgreSQL, MongoDB, MySQL
- Cloud & DevOps: AWS & GCP, Serverless Functions, Cloud Databases, API Gateways, CI/CD (Jenkins, GitHub Actions), Docker, Kubernetes, OpenShift, IAM, Infrastructure Automation
- Tools And Technologies: Git, Jira, VS Code, Postman, CI/CD Pipelines (Jenkins, GitHub Action)

ACADEMIC / EXTRA CURRICULAR ACTIVITIES

- Recognized as a "Super Node" for exemplary performance at Ongraph Technology (2022)